

Linux Server & Bash Programming — Final Exam

2025-01-07

Instructions

1. No computers, no phones, no notes
2. No talking with other people in the room
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Grade* box!)
5. For each question, you'll need to:
 - answer the question by crossing with an X the right answer
 - give an estimation of your confidence in the answer given (1-10)
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

Each question gives up to **10 points**. There are **10 questions**, for a total of maximum **100 points**.

Each correct answer gives **1 point**, each incorrect answer gives **-1 point**. The point from the answer is **multiplied by your confidence level**, for a range of **-10 to 10 points** per answer.

	Confidence 1/10	Confidence 6/10	Confidence 10/10
<i>Correct answer</i>	1 point	6 points	10 points
<i>Incorrect answer</i>	-1 point	-6 points	-10 points

Grading

Points	Grade
-100 - 50	IG
51 - 89	G
90 - 100	VG

1 In Linux everything is...

I'm /10 confident about my answer

1. Difficult
2. More fun
- ✓ 3. A file
4. A terminal
5. A shell

2 I have added new storage to my linux server. In which order do I need to create everything needed for an encrypted filesystem?

I'm /10 confident about my answer

- ✓ 1. Block device -> partitions -> LUKS encryption -> filesystem
2. Partitions -> filesystem -> block device -> LUKS encryption
3. LUKS encryption -> block device -> filesystem -> partitions
4. LUKS encryption -> block device -> partitions -> filesystem

3 How many hosts can I have in a /27 network? (NB: hosts, not addresses)

24 32

I'm /10 confident about my answer

1. 2^4
2. $2^4 - 2$
3. 2^5
- ✓ 4. $2^5 - 2$
5. 2^6
6. $2^6 - 2$

$$\begin{array}{r} 32 - \\ 27 = \\ \hline 5 \\ 2^5 - 2 \end{array}$$

net add: 00000
host add: 1111

4 How can I access another machine's localhost?

I'm /10 confident about my answer

1. By adding an entry to `/etc/hosts`
2. By creating or modifying a file in `/etc/netplan/`

- 3. It's not possible
4. By opening a listening socket in the target machine and accepting connections

5 DNS is...

I'm /10 confident about my answer

→ ① A service that translates IP addresses into hostnames

→ ② A service that translates hostnames into IP addresses

6 Bash stands for...

I'm /10 confident about my answer

1. Better Ask for Some Help

2. Better than Any other SHell

→ ③ Bourne Again SHell

4. Building Advanced Scripts Hub

7 To link together two processes, passing the output of one as the input for the next, we use...

I'm /10 confident about my answer

→ 1. |

2. >

3. >>

4. <

5. <<

⑥ 2&>

8 I'm writing a bash script, and I want to process the given cli arguments one by one, removing them from the argument list as I do so. To achieve that, I use...

I'm /10 confident about my answer

1. STDERR

2. STDIN

3. STDOUT

4. case

⑤ esac

→ 6. shift

7. unshift

9 A docker container is...

I'm /10 confident about my answer

1. Like a variable, but global
2. Like a script, but better
3. A kind of virtual machine

→ 4. An isolated process

10 In docker there are two types of volumes we can use, ...

I'm /10 confident about my answer

1. Isolated volumes and shared volumes
2. Unbound volumes and unnamed volumes

→ 3. Bind volumes and named volumes

4. It's a trick question, there is only 1 kind of volumes